

- 4 (a) r.m.s. output = $9/\sqrt{2}$ or peak input = $230\sqrt{2}$ C1
 $N_S/N_P = V_S/V_P$ C1
 $N_S = 138 \rightarrow 140$ turns A1 [3]
- (b) (i) four diodes correctly positioned regardless of output polarity M1
giving correct output polarity (*all 'point to left'*) A1 [2]
- (ii) capacitor shown in parallel with R B1 [1]
- (c) (i) time t_1 to time t_2 B1 [1]
- (ii) sketch: same peak values M1
ripple reduced and reasonable shape A1 [2]

- 5 (a) (i) ... B1 [1]